Euler's Phi-Function

- Euler's phi-function, $\phi(n)$, which is sometimes called the Euler's totient function plays a very **important role in cryptography**.
- Finds the number of integers that are both smaller than *n* and relatively prime to *n*.
- Z_n^* contains the number that are smaller than n and relatively prime to n, where as phi-function calculates the number of elements in the set.

1. $\phi(1) = 1$
2. $\phi(p) = p - 1$ if p is a prime.
3. $\phi(m \times n) = \phi(m) \times \phi(n)$ if m and n are relatively prime.
4. $\phi(p^e) = p^e - p^{e-1}$ if p is a prime.

Euler's Phi-Function

We can combine the above four rules to find the value of ϕ(n). For example, if *n* can be factored as

$$n = p_1^{e_1} \times p_2^{e_2} \times \ldots \times p_k^{e_k}$$

then we combine the third and the fourth rule to find

$$\boxed{\phi(n) = (p_1^{e_1} - p_1^{e_1-1}) \times (p_2^{e_2} - p_2^{e_2-1}) \times \cdots \times (p_k^{e_k} - p_k^{e_k-1})}$$

It is important that the calculation of phi-values for large composites can be found only if the number n can be factored into primes.

The difficulty of finding ϕ(n) depends on the difficulty of finding the factorization of *n*.

Interesting point: If *n* > 2, the value of ϕ(*n*) is even.

Example What is the value of $\phi(13)$?

Solution: Because 13 is a prime, $\phi(13) = (13 - 1) = 12$.

Example What is the value of $\phi(10)$?

Solution: We can use the third rule: $\phi(10) = \phi(2) \times \phi(5) = 1 \times 4 = 4$, because 2 and 5 are primes.

Example What is the value of $\phi(240)$?

Solution: We can write $240 = 2^4 \times 3^1 \times 5^1$. Then

$$\phi(240) = (2^4 - 2^3) \times (3^1 - 3^0) \times (5^1 - 5^0) = 64$$

Example Can we say that $\phi(49) = \phi(7) \times \phi(7) = 6 \times 6 = 36$?

Solution: No. The third rule applies when m and n are relatively prime. Here $49 = 7^2$. We need to use the fourth rule: $\phi(49) = 7^2 - 7^1 = 42$.

Example What is the number of elements in $Z_{14}*$?

Solution: The answer is $\phi(14) = \phi(7) \times \phi(2) = 6 \times 1 = 6$. The members are 1, 3, 5, 9, 11, and 13.

Fermat's Little Theorem

It's a special math shortcut that only works for PRIME numbers!

Version 1: When a is NOT divisible by p

If p is prime and a is any number NOT divisible by p, then:

$$\boxed{a^{p-1} \equiv 1 \bmod p}$$

- **In simple words:**
  Take any number a, raise it to power $(p-1)$,divide by prime $p \rightarrow$ remainder is always **1**.

Example 1:

$p = 5$ (prime)

$a = 2$ (not divisible by 5)

$2^{5-1} = 2^4 = 16$

$16 \bmod 5 = 1$

…Fermat's Little Theorem

Version 2: Works for ALL numbers a

For ANY integer a and prime p:

$$\boxed{a^p \equiv a \bmod p}$$

- It is helpful for quickly finding a solution to some exponentiations and in finding some multiplicative inverses- some times.

Example: $a = 4,\ p = 3$

LHS: $4^3 = 64$

$64 \bmod 3 = 21$ remainder 1

RHS: $4 \bmod 3 = 1$ Same remainder!

Finding Modular Inverses

The inverse of a modulo p is a^{p-2}

$$\boxed{a^{-1} \bmod p = a^{p-2} \bmod p}$$

Example: Find $4^{-1} \bmod 7$:

$4^{-1} \equiv 4^{7-2} = 4^5 \bmod 7$

$= (4^2) \bmod 7\ (4^2) \bmod 7\ (4^1) \bmod 7$

= (2x2x4) mod 7 = 16 mod 7 =2

Verify: $4 \times 2 = 8 \equiv 1 \bmod 7$

This modular inverse method works only when the modulus p is a prime number. For generalized cases we can use extended Euclidean

Example Find the result of 6^{10} mod 11.

Hint: $a^{p-1} \equiv 1 \bmod p$

Solution: We have 6^{10} mod 11 = 1. This is the **first version of Fermat's little theorem** where p = 11.

Example Find the result of 3^{12} mod 11.

Solution: Here the exponent (12) and the modulus (11) are not the same. With substitution this can be solved using Fermat's little theorem.

Hint: $a^p \equiv a \bmod p$ ***Here*** p **= 11.**

$$3^{12} \bmod 11 = (3^{11} \times 3) \bmod 11 = (3^{11} \bmod 11)\,(3 \bmod 11) = (3 \times 3) \bmod 11 = 9$$

Example **Find the remainder when 5^{123} is divided by 13. (OR find 5^{123} mod 13)**

Solution:

Step 1: Identify Prime Modulus
$p = 13$ (prime)
$a = 5$ (not divisible by 13)
Fermat's Little Theorem states: $\boxed{a^{p-1} \equiv 1 \bmod p}$

For $a = 5, p = 13$:

$$5^{12} \equiv 1 \pmod{13}$$

This reduces the exponent problem significantly.

Step 2: Break Down the Large Exponent
We need 5^{123} ,but FLT gives us $\mathbf{5^{12}} \equiv 1$.
Divide 123 by 12:

$$123 = \mathbf{12} \times 10 + 3$$

So, we can group 5^{123} into blocks of 5^{12} ,each congruent to 1.

Step 3: Apply Exponent Rules

$$5^{123} = 5^{12\times10+3} = \left(5^{12}\right)^{10} \times 5^3$$

Using FLT:

$$\left(5^{12}\right)^{10} \equiv 1^{10} \equiv 1 \pmod{13}$$

Thus:

$$5^{123} \equiv 1 \times 5^3 \pmod{13}$$

Step 4: Final Simplification

$$5^3 = 125$$

Compute $125 \mod 13$:

$$125 \bmod 13 = 8$$

So:

$$125 \equiv 8 \pmod{13}$$

Final Answer:

$$\boxed{\mathbf{8}}$$

Multiplicative Inverses

Example **The answers to multiplicative inverses modulo a prime can be found without using the extended Euclidean algorithm:**

Find $7^{-1}(\mathbf{mod 13})$

Solution:

Step 1: Verify Conditions

$p = 13$ (prime)

$a = 7$

Since , gcd(7,13)=1, the inverse exists.

Step 2: Apply Fermat's Theorem Formula

Fermat's Little Theorem for a prime p:

$$\boxed{a^{-1} \bmod p = a^{p-2} \bmod p}$$

Thus:

$$7^{-1} \equiv 7^{13-2} = 7^{11} \pmod{13}$$

Multiplicative Inverses

Solution:

Step 3: Compute $7^{11} \mod 13$ via Repeated Squaring

Break $11 = 8 + 2 + 1$:

Compute powers modulo 13:

$$7^1 = 7$$
$$7^2 = 49 \equiv 10 \pmod{13}$$
$$7^4 \equiv 10^2 = 100 \equiv 9 \pmod{13}$$
$$7^8 \equiv 9^2 = 81 \equiv 3 \pmod{13}$$

Combine:

$$7^{11} = 7^8 \times 7^2 \times 7^1 \equiv 3 \times 10 \times 7$$
$$3 \times 10 = 30 \equiv \mathbf{4} \pmod{13}$$
$$\mathbf{4} \times 7 = 28 \equiv 2 \pmod{13}$$

So:

$$7^{-1} \equiv 2 \pmod{13}$$

Step 4: Verification

$$7 \times 2 = 14 \equiv 1 \pmod{13}$$

Confirmed.

Practice Exercise :

1. Find 3^{64} mod 41
2. Find $9^{-1}(mod 19)$
3. Find $12^{-1}(mod 29)$

Practice Exercise 1 solution:

1. Find 3^{64} mod 41

Apply Fermat's Little Theorem
Since gcd(3, 41) = 1, $3^{40} \equiv 1 \pmod{41}$.

Reduce the exponent
$3^{64} = 3^{40} \times 3^{24}$
$3^{64} \equiv 3^{24} \pmod{41}$.
Compute 3^{24} mod 41
$3^2 = 9$
$3^4 = 81 \equiv -1 \pmod{41}$
$3^8 = (-1)^2 \equiv 1 \pmod{41}$
$3^{16} \equiv 1 \pmod{41}$
$3^{24} = 3^{16} \times 3^8 \equiv 1 \times 1 \equiv 1 \pmod{41}$
Final Answer:
$3^{64} \equiv 1 \pmod{41}$

Euler’s Theorem

Think of it as Fermat's Theorem for ANY number!

Fermat's Theorem: Works only for Modulus is **prime number and** base & modulus are coprime

Euler's Theorem: Works for Modulus is **any positive number and** base & modulus are coprime

Version 1: When a and n are co-prime

If , then:

$$\boxed{a^{\phi(n)} \equiv 1 \pmod{n}}$$

In simple words:

Take any number a that shares no factors with n, raise it to power $\phi(n)$,divide by $n \rightarrow$ remainder is **1**.

Euler's Theorem

Version 2: The RSA Connection

If $n = p \times q$ (product of two primes), then:

$$\boxed{a^{k \times \phi(n) + 1} \equiv a \pmod{n}}$$

k is an integer

This is the mathematical heart of RSA encryption!

- Used to secure internet communications
- Makes online banking and messaging private

Example **Find the remainder when 2^{50} is divided by 11.**

Solution:

Step 1: Verify Conditions

$n = 11$

$a = 2$ (coprime)

Euler's theorem applies .

Step 2: Apply Euler's Totient Function

For a prime p:

$$\phi(p) = p - 1$$

So:

$$\phi(11) = 10$$

Step 3: Use Euler's Theorem Formula

Euler's theorem states:

Thus:

$$2^{10} \equiv 1 \pmod{11}$$

Step 4: Simplify the Large Exponent

We want $2^{50} \mod 11$.

$2^{50} = \left(2^{10}\right)^5$

Step 5: Apply Euler's Result

$$(2^{10})^5 \equiv 1^5 \equiv 1 \pmod{11}$$

Therefore:

$$2^{50} \equiv 1 \pmod{11}$$

Final Answer: Remainder when 2^{50} is divided by 11 is $\boxed{1}$

Multiplicative Inverses

Euler's theorem can be used to find multiplicative inverses modulo a composite. If n and a are coprime, then

$$\boxed{a^{-1} \bmod n = a^{\phi(n)-1} \bmod n}$$

Example Find $5^{-1}(\mathbf{mod 18})$

Step 1: Verify Conditions

$n = 18$ (composite)

$a = 5,$ (coprime)

Inverse exists since a and n are coprime.

Step 2: Compute Euler's Totient $\phi(n)$

Prime factorization:

$$18 = 2 \times 3^2$$

Thus:

$$\phi(18) = \phi(2)x\phi(3^2) = 1\text{x}(3^2 - 3^1) = 6$$

Step 3: Apply Euler's Theorem Formula

$$a^{-1} \equiv a^{\phi(n)-1} \pmod{n}$$

For $a = 5$, $n = 18$, $\phi(18) = 6$:

$$5^{-1} \equiv 5^{6-1} = 5^5 \pmod{18}$$

Multiplicative Inverses

Euler's theorem can be used to find multiplicative inverses modulo a composite. If n and a are coprime, then

$$\boxed{a^{-1} \bmod n = a^{\phi(n)-1} \bmod n}$$

Example

Find $5^{-1}(\mathbf{mod18})$

Step 4: Compute $5^5 \bmod 18$

$$5^2 = 25 \equiv 7 \pmod{18}$$

$$5^4 = \left(5^2\right)^2 \equiv 7^2 = 49 \equiv 13 \pmod{18}$$

$$5^5 = 5^4 \times 5 \equiv 13 \times 5 = 65 \equiv 11 \pmod{18}$$

Thus:

$$5^{-1} \equiv 11 \pmod{18}$$

Verification

$$5 \times 11 = 55$$

$$55 \div 18 = 3 \text{ } \mathit{remainder} \text{ } 1$$

$$55 \equiv 1 \pmod{18}$$

Final Answer: $\boxed{11}$

Practice Exercise :

1. Remainder when 5^{100} is divided by 17.
2. Remainder when 3^{75} is divided by 13.
3. Remainder when 4^{40} is divided by 9.
4. Find $17^{-1}(mod 33)$
5. Find $13^{-1}(mod 40)$

Practice Solution 1:

Remainder when 5^{100} is divided by 17. (OR find 5^{100} mod 17)

Step 1: $n = 17$, prime, $\gcd(5,17) = 1$.

Step 2: $\phi(17) = 16$.

So $5^{16} \equiv 1 (\mathrm{mod} 17)$.

Step 3: Exponent division

$100 \div 16$: quotient 6, remainder 4.

So $100 = 16 \times 6 + 4$.

Therefore:

$$5^{100} = 5^{16\times 6} \times 5^4 = (5^{16})^6 \times 5^4.$$

Step 4: Using Euler:

$5^{16} \equiv 1$, so $(5^{16})^6 \equiv 1^6 \equiv 1$.

So $5^{100} \equiv 1 \times 5^4 (\mathrm{mod} 17)$.

Step 5: Compute 5^4 mod 17

$5^2 = 25, 25\, mod\ 17 \equiv 8$ mod 17

$5^4 = 8\, x\, 8 = 64, 64\, mod\ 17 \equiv 13$ mod 17.

So remainder = **13**.